Junior Mathematical Olympiad – 2026

Answer all questions. Give Justification to your answer.
Use of calculator (in any form) is not allowed.
(All questions carry equal marks)

Time: 3 hours　　　　　　　　　　　　　　　　　　　　Full Mark: 100

1. Find the value of $\frac{(4\times7+2)(6\times9+2)(8\times11+2)\ldots\ldots\ldots(100\times103+2)}{(5\times8+2)(7\times10+2)(9\times12+2)\ldots\ldots\ldots(99\times102+2)}$

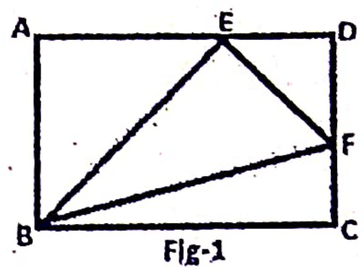

Fig-1

2. In the given figure(Fig-1) ABCD is a rectangle. E & F are mid points of AD and CD respectively. If area of ΔBEF is 9 sq. cm. Find the area of rectangle ABCD.
3. A three-digit number is equal to 16 times of the sum of its digits and equal to 9 times of the product of its digits. Find the number.
4. Find the least number, which when divided by 243, the remainder is 146 and when it is divided by 247, the remainder is 98. Find the number.
5. How many three-digit numbers can be written using both odd and even digits in each and every number ?
6. When Swati cleaned the Almirah racks the 'Old Mahabharat Book' fell from her hand. Several 30 sheets fell from the book. There are two pages in each sheet. Then could the sum of the page numbers be 6000(six thousand). Answer with reason.
7. In group of birds, one-fourth of them are seating on the tree, one-fifth of them are inside the nest and the remaining 22 birds of the group are in the field. Find the number of birds inside the nest ?
8. A metallic cuboid is formed by melting 3 cuboids of sides 6 cm, 8 cm and 10 cm. Find the length of side of the new cuboid.

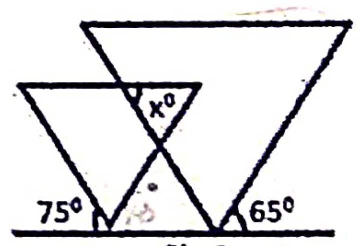

Fig-2

9. In the given figure(Fig-2), the two triangles are equilateral triangles. Find the value of x.
10. Find the sum of all digits in the result obtained by simplification of $(10^{99}-99)$.
11. In an examination paper there are 30 questions. One gets 5 marks for each correct answer and 2 marks are deducted for each wrong answer. If one gets total of 122 marks in this examination, find how many correct answers and how many wrong answers are written by him ?
12. 20 years back, a man was 5 times as old as his son and after 16 years the age of the man will be 41 years. What is the present age of the man ?
13. The sum of 5 consecutive odd numbers is 1185. Find these numbers.
14. In a 3-digit number, 6 is added to 100th place and 6 is subtracted from unit place then new number is 3 times of given number. Find the given number.
15. Find the last two digits of $2^{6n}-6^{2n}$ in which n is a natural even number.
16. ABCD is a parallelogram in which AB=7 cm. and diagonal AC=8 cm. BE is perpendicular to AC. If BE=$3\sqrt{5}$ cm. Find the length of the diagonal BD.
17. In a school, there are only two courses, mathematics and science. Ratio of boys and girls in the school is 5:4. But number of boys and girls in mathematics are equal. If number of mathematics students are 25% more than the science students, find the ration of boys and girls in science subject.

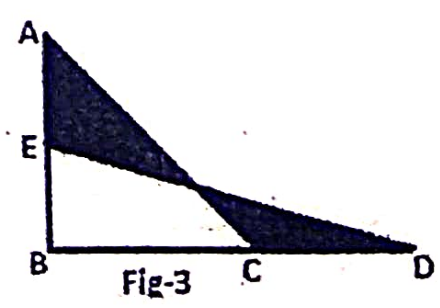

Fig-3

18. In the given figure(Fig-3) AB=BD=12 cm, BE=BC=5 cm and $m\angle B = 90°$. Find the area of the shaded region.
19. A man went out between 4 pm and 5 pm. He returned home between 7 pm and 8 pm. He found that the minute hand and the hour hand exactly changed their places. When did he go out of his home ? and how much time did he spend out side ?
20. Perimeter of a rhombus is 40 cm and sum of its diagonals is 28 cm. Find the area of that rhombus.

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